

Flight Risk Phase 2 Report

Since phase one, the flight risk project has successfully transitioned from a conceptual framework into a functional Prototype. We moved our raw data and plan to an MVP interface designed to mitigate the uncertainties of domestic commercial air travel. While phase one identifies the concept of this idea, phase two puts it into a working prototype that predicts different flight outcomes. The prototype we created synthesizes historical weather patterns, allowing it to move beyond binary delay predictions. The current state of our prototype utilizes a gradient classification model to quantify specific probabilities for multiple distinct outcomes.

This prototype demonstrates substantial technical progress; we have successfully pulled historical data with weather features and flight patterns, enabling our model to account for environmental disruptions and patterns based on flight carriers. The model produces five outcomes: on time, short delay, long delay, canceled, and it provides the lowest risk flight. Users also have a choice of how many flight options they would like to view. Using codex, enabled our model to create a risk scoring algorithm that implements a weighted risk score formula to calculate the outcome for each category. This model implements a very user-friendly dashboard where users can filter flights by inputting where they are flying from and to, and their preferred airline carrier, which then produces a ranked list of options for a specific flight date and route. Instead of just showing percentages or a yes/no feature, our prototype generates a natural language guide that makes it easy to read for users and very accessible to non-tech users. It also uses a probabilistic ranking system by calculating specific probabilities for the four categories, allowing us to differentiate flights that would be 5 min late from one that has a high cancellation rate. For example, if a flight produces a 6.5% risk score, the model outputs historical data that had a lower distribution profile or an elevated risk due to weather.

Currently, our top three technical risks and open bottlenecks are timely data, irregular operations, and changing operations. Regarding timely data, the historical flight data from 2018 to 2022 was easy to obtain, but the current or live data is harder to access. Older data may be less reliable for modern flight predictions. Irregular operations, extreme storms, strikes, and other unusual disruptions may make the algorithm less reliable because these outlier events do not appear often enough in the training data. Finally, regarding changing operations, airlines and airports change their operating patterns over time, meaning our model will eventually need regular updates or pipeline automation to remain accurate.

There are still elements that do not yet work and will be completed before Phase three. For weather integration, we have merged weather data from the same time

period with the flight data. We are going to try to implement more current data. Additionally, our user interface and ranking logic take a user's input and output a ranked list of flights that are not fully implemented yet. We also plan to add a budget feature that produces flight prices as a category when the program is ran to show users the least at-risk options and the prices associated with them based on different airline companies. We also want to make this result a user-friendly statement that will help guide their decision.

Below are a few screenshots of how the code responds when it's given a few different requests:

Flight Risk

Prototype flight disruption ranking tool built from historical flight and weather data.

Origin airport

Destination airport

Travel date

LAX

JFK

2026/04/28

Airline filter

Any

How many options to show

10

Rank Flight Options

flight_code	airline_code	departure_time	distance	scheduled_elapsed_time	risk_score	on_time	short_delay	long_delay	cancelled	historical_support	explanation
DL392	DL	9:00 AM	2475	345	3.7%	82.6%	16.9%	0.3%	0.2%	2	historically lower disruption profile
DL324	DL	7:00 AM	2475	323	4.4%	79.4%	20.0%	0.4%	0.2%	2	historically lower disruption profile
DL324	DL	7:15 AM	2475	311	5.0%	76.9%	22.5%	0.4%	0.2%	1	historically lower disruption profile
DL319	DL	10:15 AM	2475	312	5.3%	74.9%	24.6%	0.3%	0.2%	1	historically lower disruption profile
DL328	DL	9:59 PM	2475	331	5.4%	75.3%	23.7%	0.9%	0.1%	1	historically lower disruption profile
AA2	AA	8:00 AM	2475	340	5.4%	78.6%	19.4%	1.0%	1.0%	2	historically lower disruption profile
DL2521	DL	11:59 PM	2475	335	5.5%	74.8%	24.3%	0.8%	0.1%	1	historically lower disruption profile
AA238	AA	10:00 AM	2475	342	5.5%	77.8%	20.4%	0.9%	0.9%	2	historically lower disruption profile
DL323	DL	1:00 PM	2475	342	5.9%	74.4%	23.8%	1.6%	0.1%	1	historically lower disruption profile
AA28	AA	7:30 AM	2475	318	6.3%	74.7%	23.3%	1.0%	1.1%	1	historically lower disruption profile

Lowest-risk option: DL at 09:00 with risk score 3.7%.

This shows a flight from LAX (California) to JFK (New York) - it shows 10 flight options with several different carriers, along with the four categories and lowest risk flight available.

Flight Risk

Prototype flight disruption ranking tool built from historical flight and weather data.

Origin airport

PHX

Destination airport

DEN

Travel date

2026/04/28

Airline filter

Any



Rank Flight Options

flight_code	airline_code	departure_time	distance	scheduled_elapsed_time	risk_score	on_time	short_delay	long_delay	cancelled	historical_support	explanation
AA2511	AA	10:34 AM	602	121	5.6%	77.8%	20.3%	0.8%	1.0%	1	historically lower disruption profile
AA1332	AA	9:36 AM	602	117	5.7%	77.6%	20.5%	0.8%	1.1%	1	historically lower disruption profile
AA1332	AA	9:17 AM	602	112	6.2%	76.1%	21.7%	0.8%	1.4%	1	historically lower disruption profile
UA1253	UA	6:15 AM	602	113	6.4%	72.9%	25.4%	0.9%	0.7%	1	historically lower disruption profile
F9750	F9	6:00 AM	602	118	6.4%	73.7%	24.5%	0.9%	1.0%	1	historically lower disruption profile
UA604	UA	9:40 AM	602	113	6.6%	70.9%	27.7%	0.8%	0.5%	1	historically lower disruption profile
AA897	AA	6:02 AM	602	106	6.6%	76.3%	20.9%	0.9%	1.9%	1	historically lower disruption profile

Lowest-risk option: AA at 10:34 with risk score 5.6%.

This is a flight from PHX (Arizona) to DEN (Colorado) - this shows 7 flight options with different airline carriers along with the four categories and the lowest risk flight available.

Flight Risk

Prototype flight disruption ranking tool built from historical flight and weather data.

Origin airport

MSP

Destination airport

ATL

Travel date

2026/04/28

Airline filter

Any



Rank Flight Options

flight_code	airline_code	departure_time	distance	scheduled_elapsed_time	risk_score	on_time	short_delay	long_delay	cancelled	historical_support	explanation
DL1744	DL	6:20 AM	907	148	2.8%	87.2%	12.3%	0.3%	0.1%	1	historically lower disruption profile
DL437	DL	5:05 AM	907	158	2.8%	88.8%	10.0%	1.1%	0.2%	1	historically lower disruption profile
DL1770	DL	7:52 AM	907	146	2.9%	86.7%	12.9%	0.3%	0.1%	1	historically lower disruption profile
DL1770	DL	7:30 AM	907	146	2.9%	86.5%	13.0%	0.3%	0.1%	1	historically lower disruption profile
DL2243	DL	9:00 AM	907	151	3.0%	86.1%	13.5%	0.3%	0.1%	1	historically lower disruption profile
DL320	DL	10:15 AM	907	149.5	3.1%	85.5%	14.1%	0.3%	0.1%	2	historically lower disruption profile
DL1048	DL	11:03 AM	907	147	3.3%	84.4%	15.2%	0.3%	0.1%	2	historically lower disruption profile
NK1561	NK	8:37 AM	907	158	4.4%	81.3%	17.4%	0.9%	0.4%	1	historically lower disruption profile
DL1081	DL	12:55 PM	907	145.5	4.8%	79.6%	18.7%	1.6%	0.1%	2	historically lower disruption profile
DL434	DL	12:59 PM	907	145	4.8%	79.6%	18.8%	1.6%	0.1%	1	historically lower disruption profile
DL434	DL	1:00 PM	907	145	4.8%	79.6%	18.8%	1.6%	0.1%	1	historically lower disruption profile

Lowest-risk option: DL at 07:35 with risk score 2.8%.

This is a flight from MSP(Minesota) to ATL (Georgia) - this shows 15 flight options with different airline carriers along with the four categories and the lowest risk flight available.